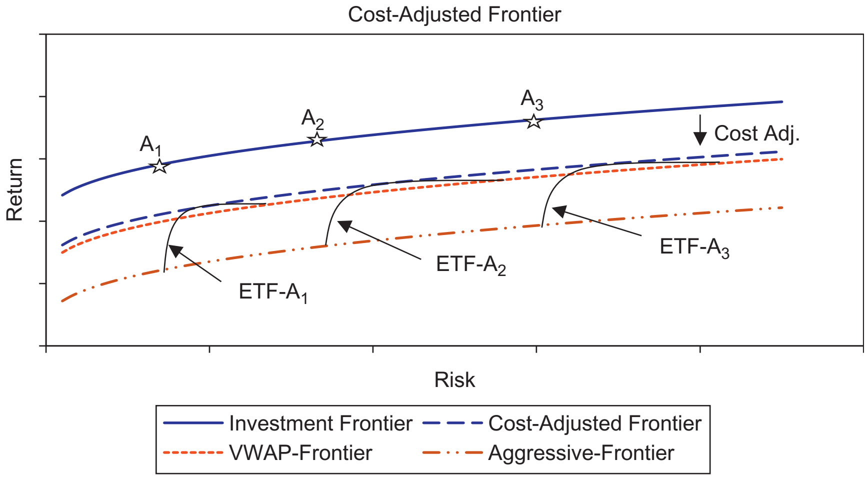

■ **Figure 10.6** Cost Adjusted Frontier. *Source: Journal of Trading (2007).*

and normal strategies. Then, the best the investor can achieve is the envelope of the highest cost-adjusted points. This frontier is now referred to as the cost-adjusted frontier.

The VWAP frontier shows the expected risk-return profile for the optimal portfolios if VWAP was used to implement the decision. The aggressive frontier shows the expected risk-return profile for the optimal portfolios if an aggressive strategy was used to implement the decision. The optimal cost-adjusted frontier is the upper envelope of all the cost-adjusted portfolios.

It is interesting to point out here that in our example, the VWAP strategy is an inefficient ex-post frontier because the VWAP frontier lies below the cost-adjusted frontier and it is associated with a lower level of investor utility.

Another interesting aspect is that the VWAP frontier is equivalent to a cost-adjusted frontier constructed from an aggressive strategy (aggressive frontier). Notice that the VWAP frontier passes through the most passive strategies on the efficient trading frontier as well as a more aggressive strategy on the efficient trading frontier. Neither the passive VWAP frontier nor the aggressive frontier is a preferred strategy since they do not maximize investor utility. Therefore, execution via a VWAP or overly aggressive strategy leads to decreased utility. To maximize utility it is essential that the underlying trading strategy not incur too much cost (primarily market impact) or too much risk.